

TEAM 3

**SMART
WASTE MANAGEMENT
SYSTEM**

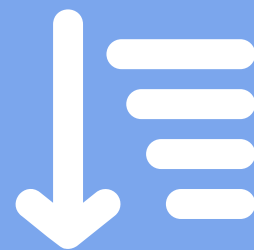
Overflowing Urban Waste Bins



Overflowing Urban Waste Bins



Urban Hygiene &
Visual Degradation



Inefficient Waste
Collection



Lack of Operational
Visibility



Name : Andy (35 years old)

Occupation : District Operations Manager

Housing type : Small apartment near Bellevue

Income level : Middle class

Responsibilities

- Regularly collect household waste bins within designated areas
- Clean up overflowing waste and surrounding areas during collection
- Visit an average of 80 to 120 bins per day

Pain Points

Lack of prior information: All bins visited under the same criteria (inefficient), approximately 35% of daily collection points were in a state where collection was unnecessary.

Workload Increase: Stress escalates due to exceeding planned time and hygiene issues caused by cleaning up overflowing trash.

Physical strain: Energy expenditure due to unnecessary movement

Needs

I want to selectively visit only the places where collection is truly necessary.

Collect before overflowing to minimize additional cleanup work

Minimize unnecessary movement to efficiently manage daily work routes

Premium Garbage Collection Subscription Service



Garbage can
sensor attached



Real-time
monitoring



Automatic
Collection



Overflow
Prevention



Reduced burden of
user management



Improved collection
efficiency

Reasons for Idea Failure

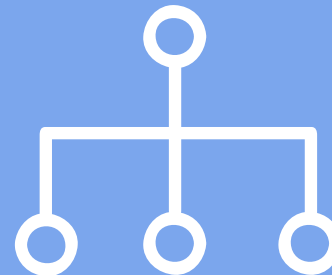


Unsuitable for the operating environment



An environment where trash cans are not placed outside

Cost structure issues



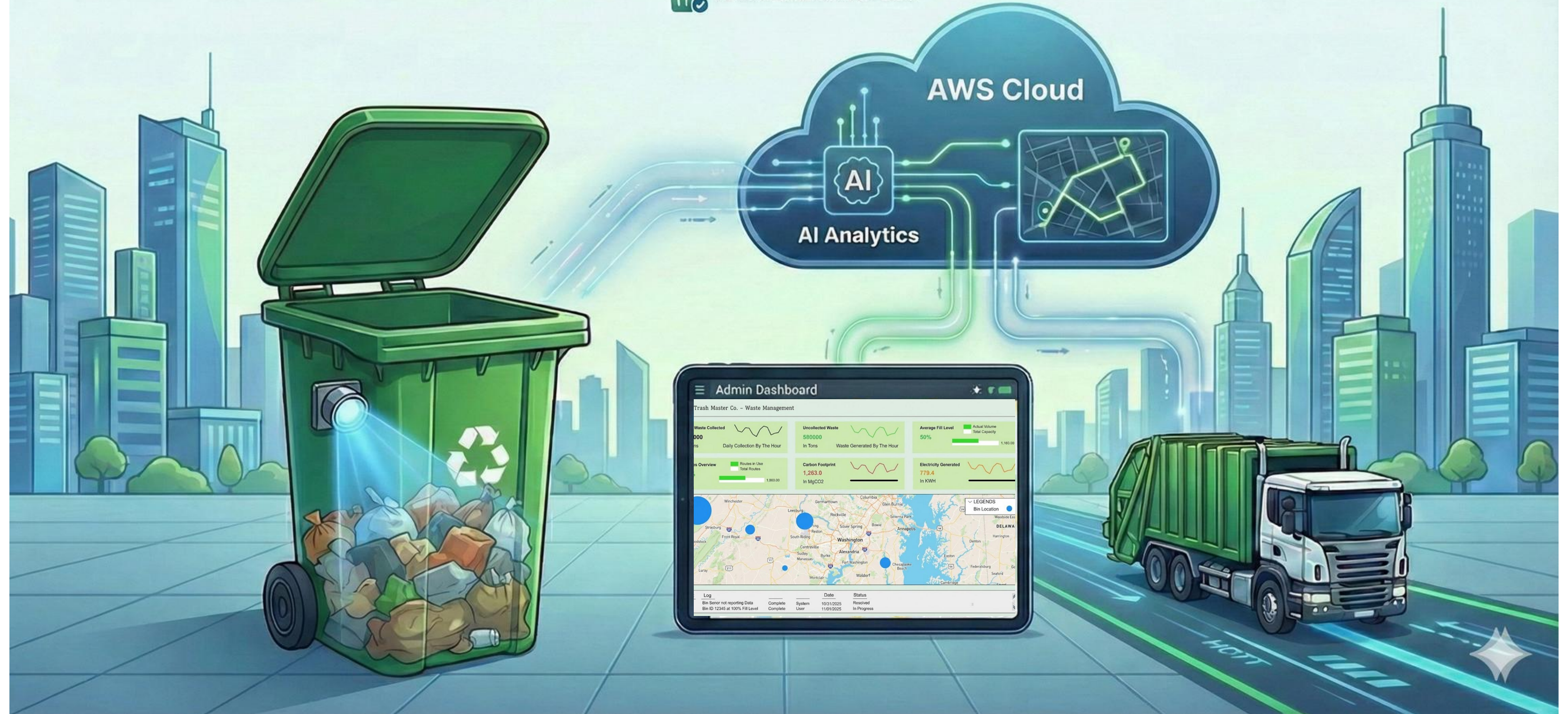
In addition to public collection fees, a structure requiring an additional subscription fee

Structural deficit

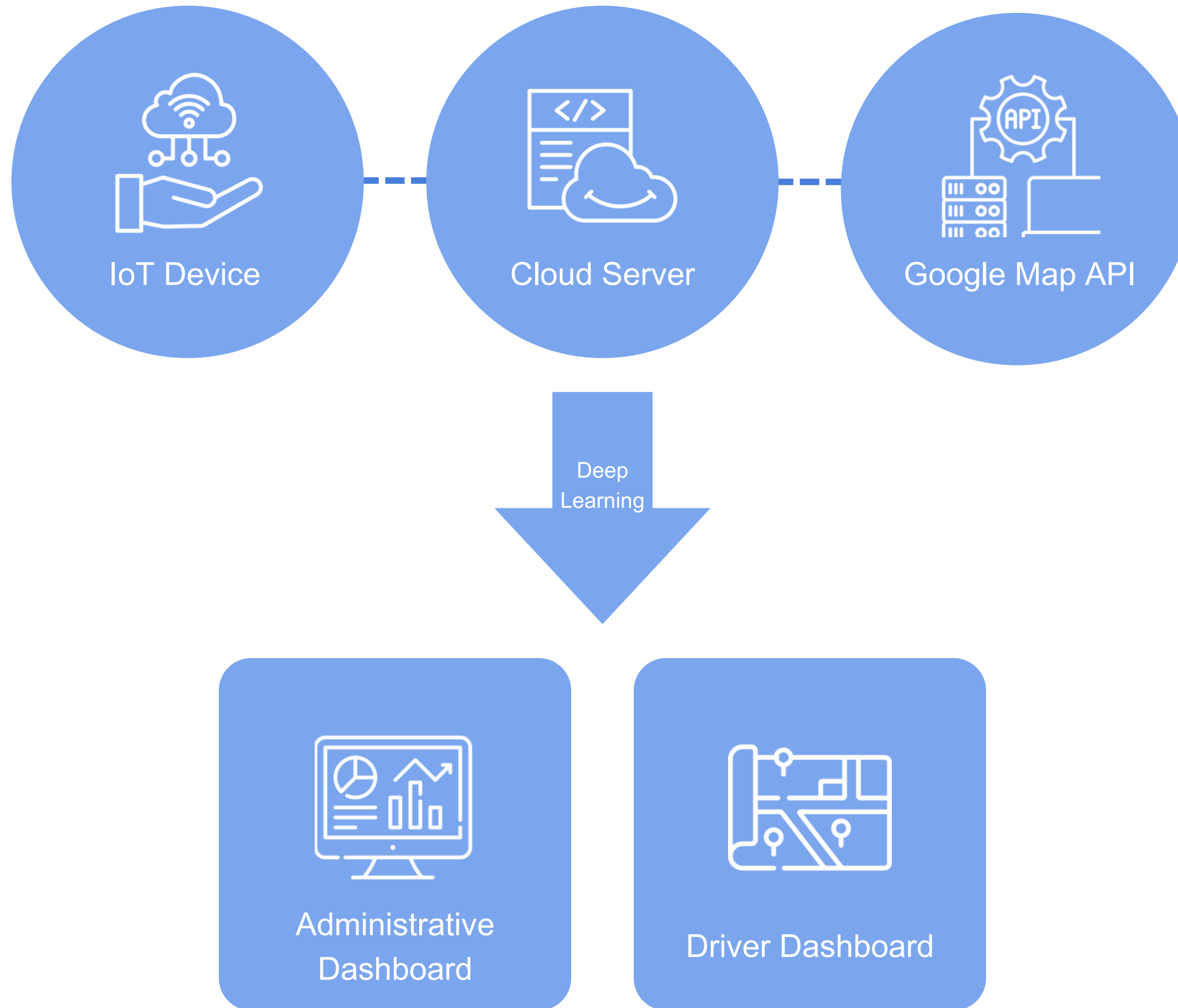


If sufficient users are not secured, it will result in a deficit.

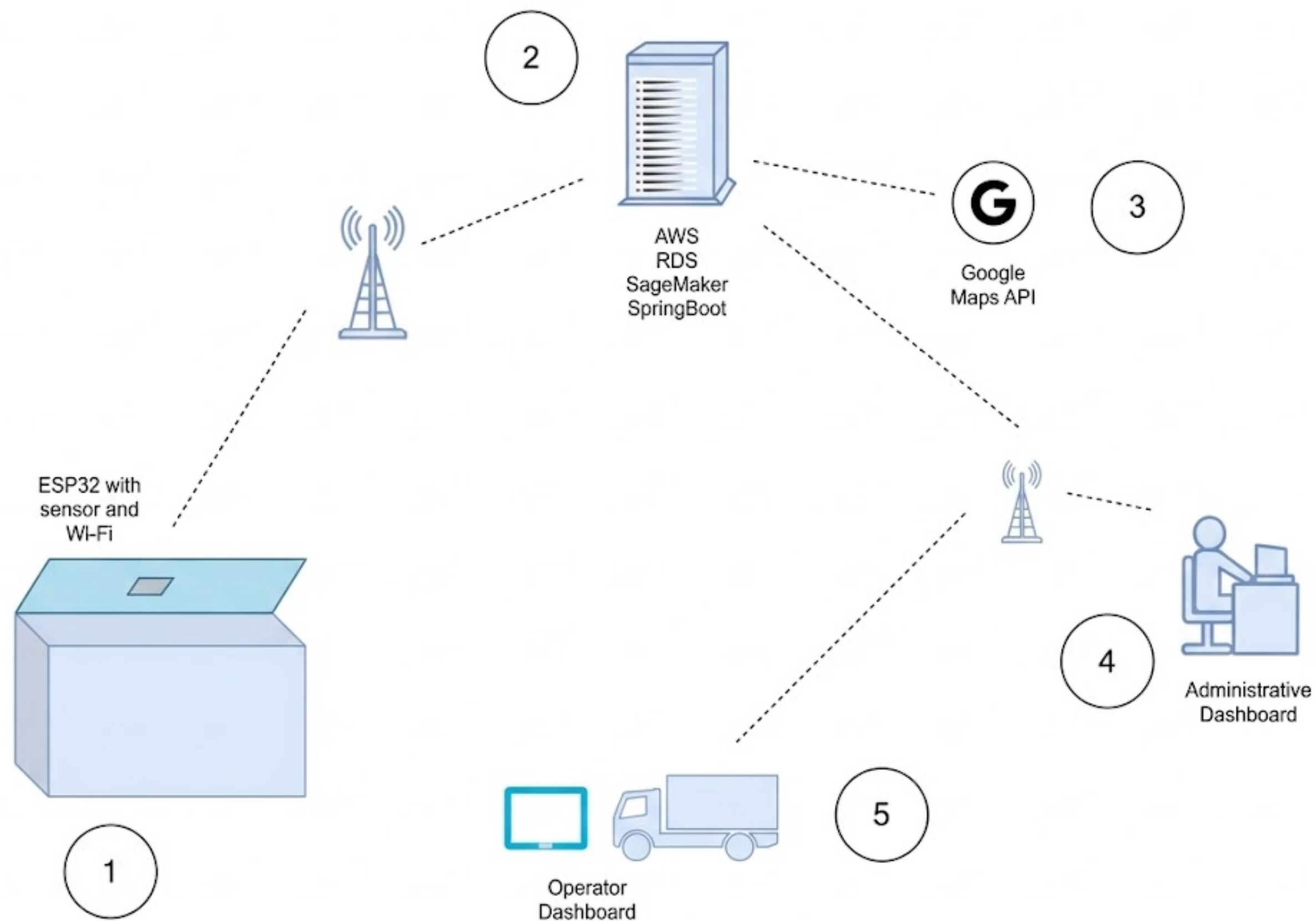
Smart Waste Management System



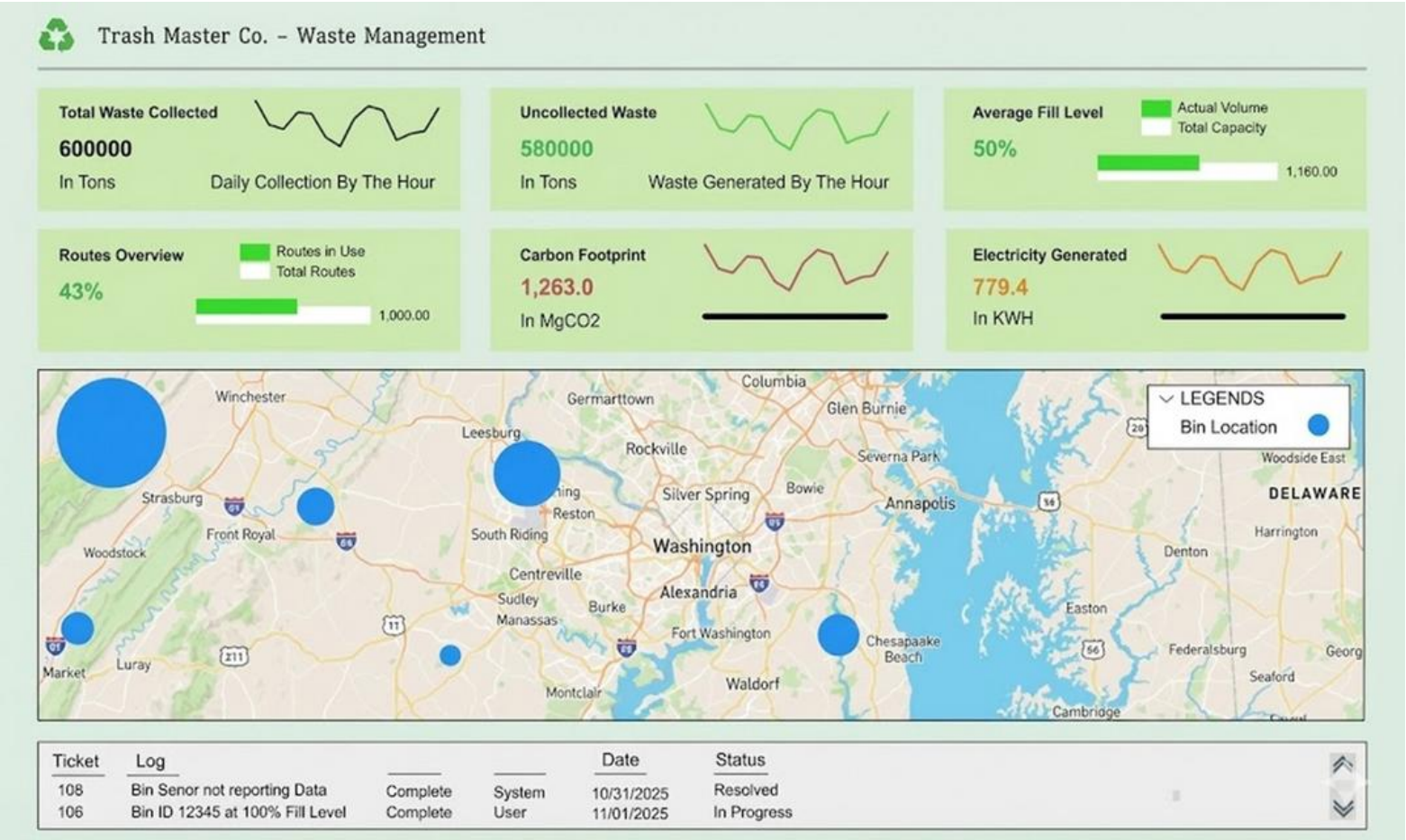
part 2
Solution



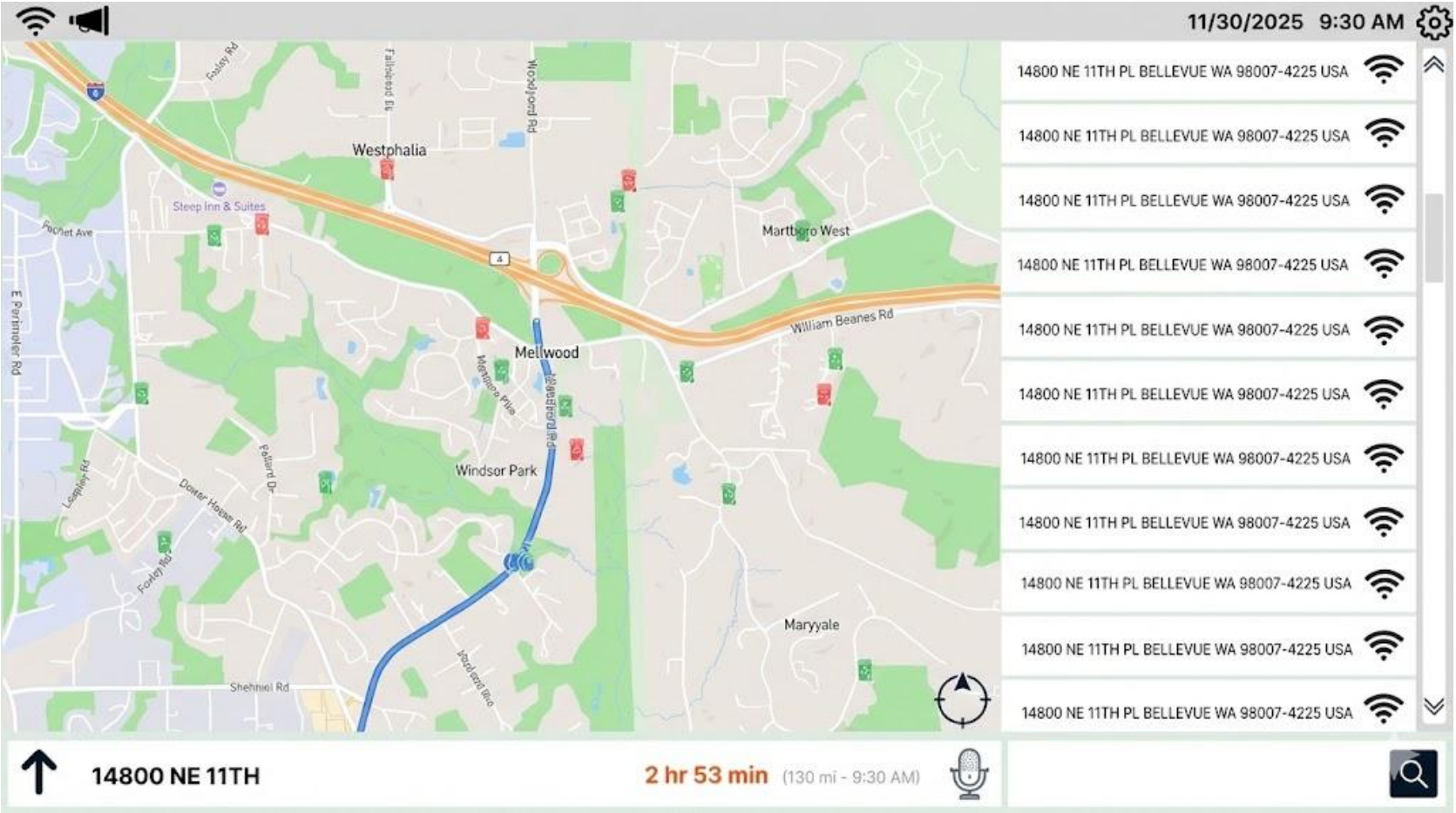
part 2 Solution



part 3
impact



Administrative Dashboard



Driver Dashboard

1

People living in Seattle must be feeling inconvenienced by the overflow of trash.

2

Using our product will improve waste collection efficiency.

3

If our product can improve efficiency, city governments and waste collection companies will use it.

1

People living in Seattle must be feeling inconvenienced
by the overflow of trash.

Micro User Feedback

Location

Location: Seattle/Bellevue public spaces

Duration: 1–2 days

Method Type: Problem validation
(before solution)

Micro Feedback

Short, informal one-sentence questions
(≤30 seconds)

Target: Citizens

Question:

“Does overflowing trash here bother
you?”

Record: Yes / No, Short comment

Validation Criteria

The hypothesis is supported if:
Majority of respondents express
inconvenience or concern

2

Using our product will improve waste collection efficiency.

Instead of actual sensors, 'human sensors' repeatedly visit the same trash bins at set times, manually grading and recording photos + load levels to simulate real-time fill level data.

Trash Bin Selection

Designate public trash bins and mark their locations on the map.

Recording load capacity with a 'human sensor'

Visit the same trash bin 2-3 times at regular intervals to visually record the load volume, noting the time and taking photos.

Full Collection vs. Selective Collection

Calculate and compare the distance/time for each method using Google Maps

3

If our product can improve efficiency,
city governments and waste collection companies will use it.

Short User Reaction Test

Participants

- Campus facilities staff
- Cleaning staff



Scenario Shown

“Check bin fill levels
before collection”

Key Questions

- Unnecessary pickups?
- Would this improve efficiency?
- Would you try it as a pilot?

Economic Efficiency



Reduce empty pickups
Optimize fuel and labor
costs

Proactive Urban Management

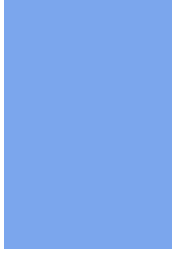


Prevent bin overflows
Improve hygiene and public
satisfaction

Digital Transformation



Data-driven decision-making
Scalable waste management
system



Q&A